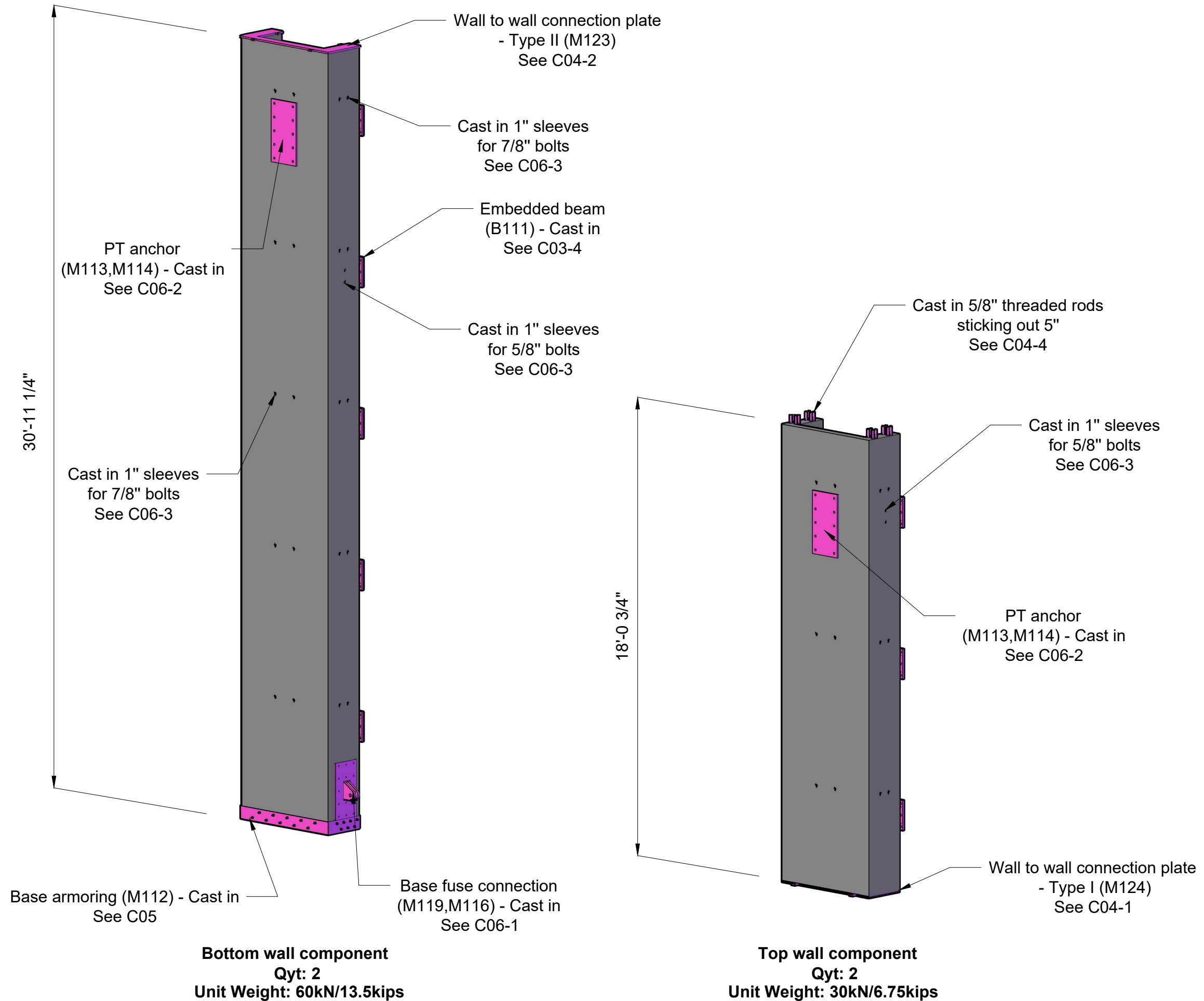
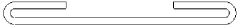
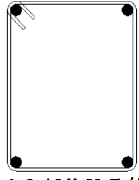
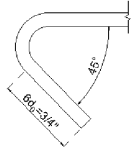
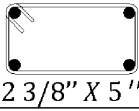




Rebar Information

Name	Shape	Size	Anchorage Type	Unit Length	Qty	Supplier
Vertical zone reinforcement -Top wall		#3 3/8"	/	18'	64	LMS (UBC bends all the ties, LMS bends the horizontal rebar)
Vertical zone reinforcement -Bottom wall				31'	64	
Vertical distributed reinforcement - Top wall				18'	48	
Vertical distributed reinforcement - Bottom wall				31'	48	
Horizontal distributed reinforcement at flanges		#3 3/8"		2' - 1"	784	
Horizontal distributed reinforcement at web				5' - 1"	392	
Ties in zone (undeformed wire) Type I	 6 3/8" X 5"			2'	1622	
Ties in zone (undeformed wire) Type II	 2 3/8" X 5"			1' - 4"	1622	
Additional wires at flanges	 4 7/8"			7 1/2"	300	
Additional wires at webs	 1 3/8"			5"	430	
Threaded Dywidag bar		Ø5/8"	/	2' - 3"	16	Dywidag
Material Summary						
Type	Total length	Volume [inch3]		Weight [kg]		
#3 deformed rebar	9114 ft	12073		1543		
11ga undeformed wire	5773 ft	783		100		
Threaded Dywidag bar	36 ft	132		17		

Sleeve Information

Inner Diameter	Unit Length	Qty	Total length
Ø1"	6"	64	32 ft
Ø3/4"	6"	16	8 ft
Ø1"	3 1/2"	32	9.3 ft

Material Properties

Unless otherwise specified in the drawings, materials for all components should be:

Rebar and Wire:
#3 rebar : ASTM 706 Grade 60

11ga wire: Grade ASTM A1064

Concrete:
f 'c = 45 Mpa, from 28 day concrete cylinder tests

Clear cover is 5/16"

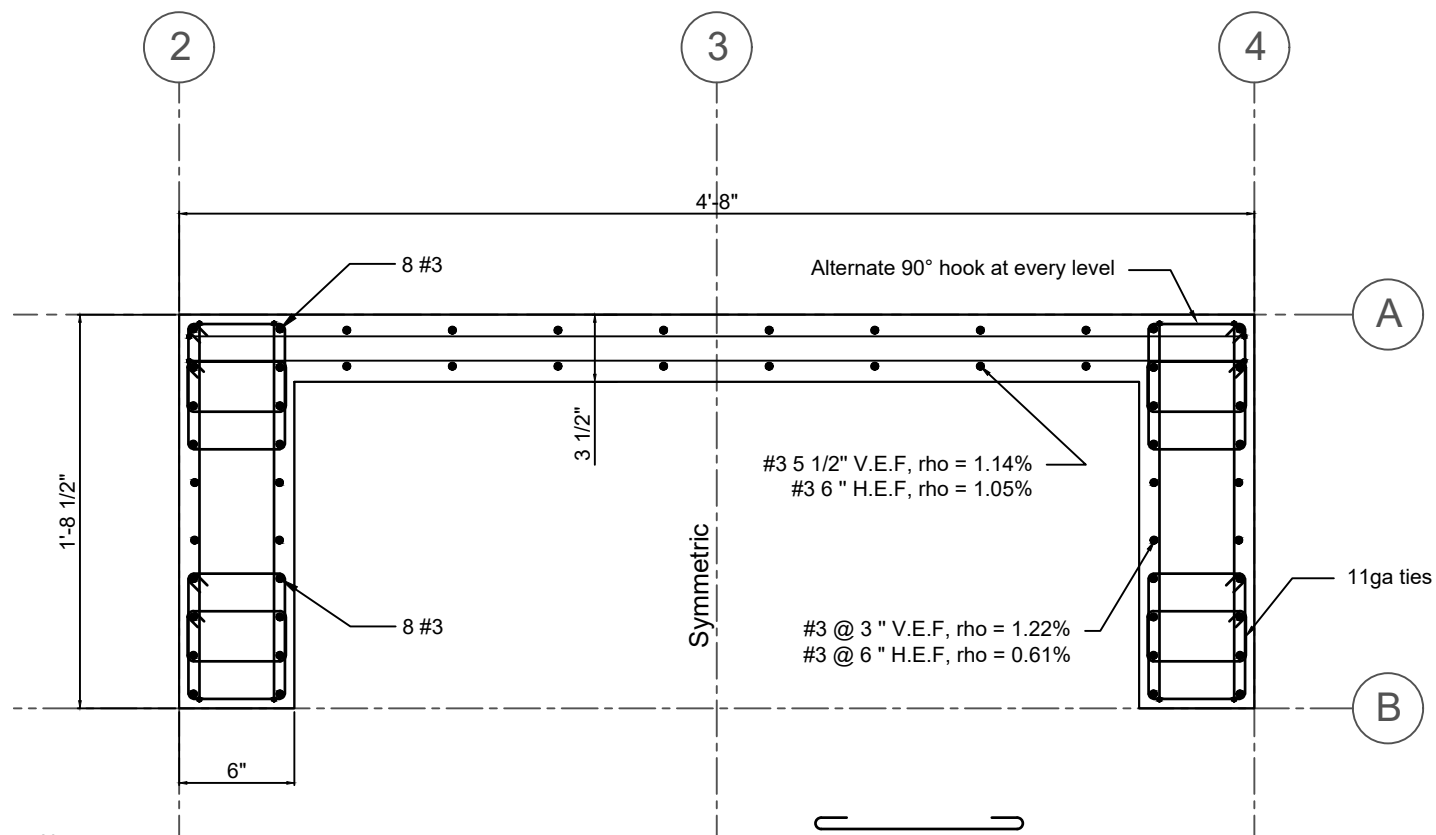
Max. aggregate size is 1/4"

Cast in Steel Components

M112 - Base armoring
M113, M114 - PT anchor
M116, M119 - Base fuse connection
M123 - Wall to wall connection plate Type II
M124 - Wall to wall connection plate Type I
B111 - Embedded beam

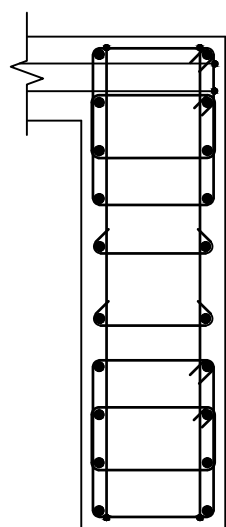
Volume of Wall Piers

7.704 m^3 (Four pieces together)

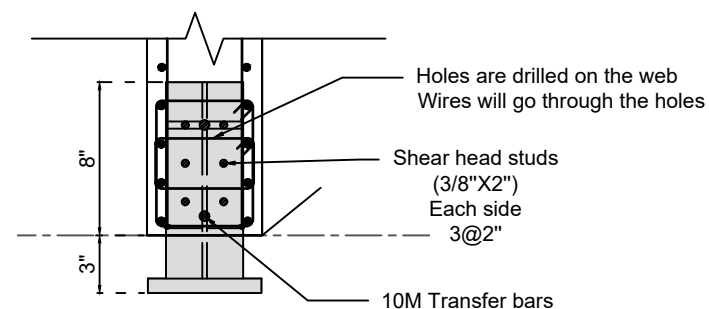


Note:
#3 : dia = 3/8"
11ga : dia = 0.12"
Minimum clear thickness 5/16"

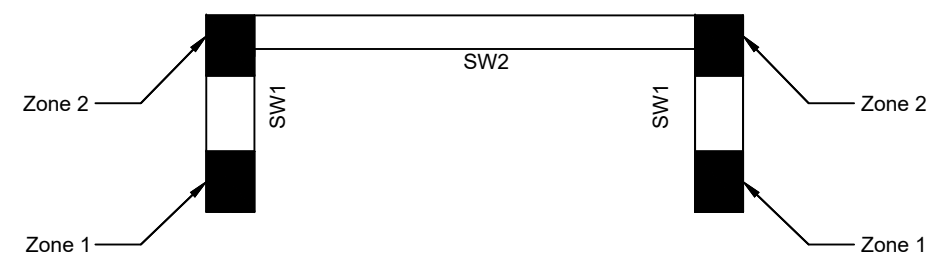
Hook horizontal bars 180° both ends
Hook tight to end of zone at end of wall
(Alternate 90° hook if necessary)



Additional hoops @2 1/4"
between level GND and L2



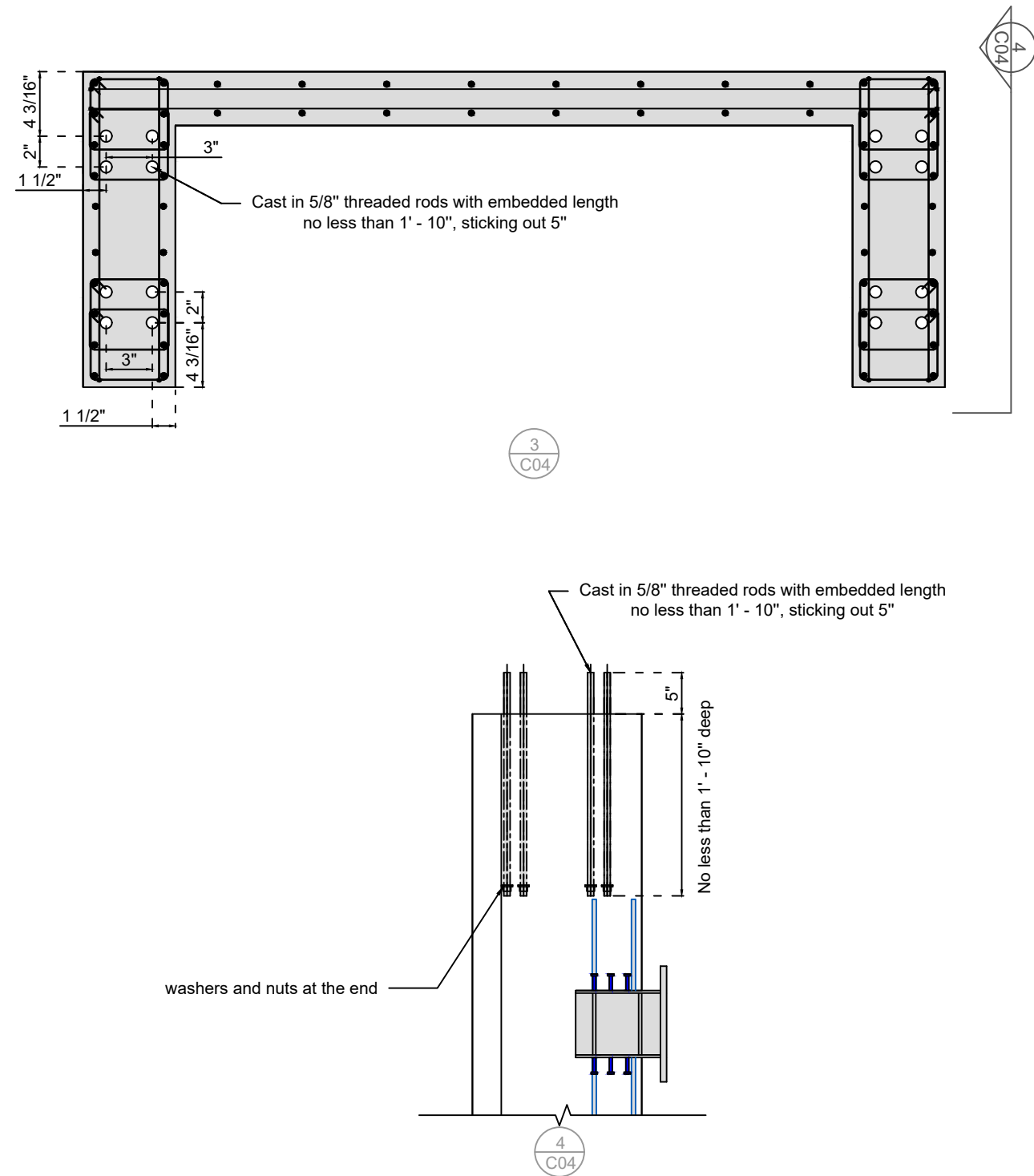
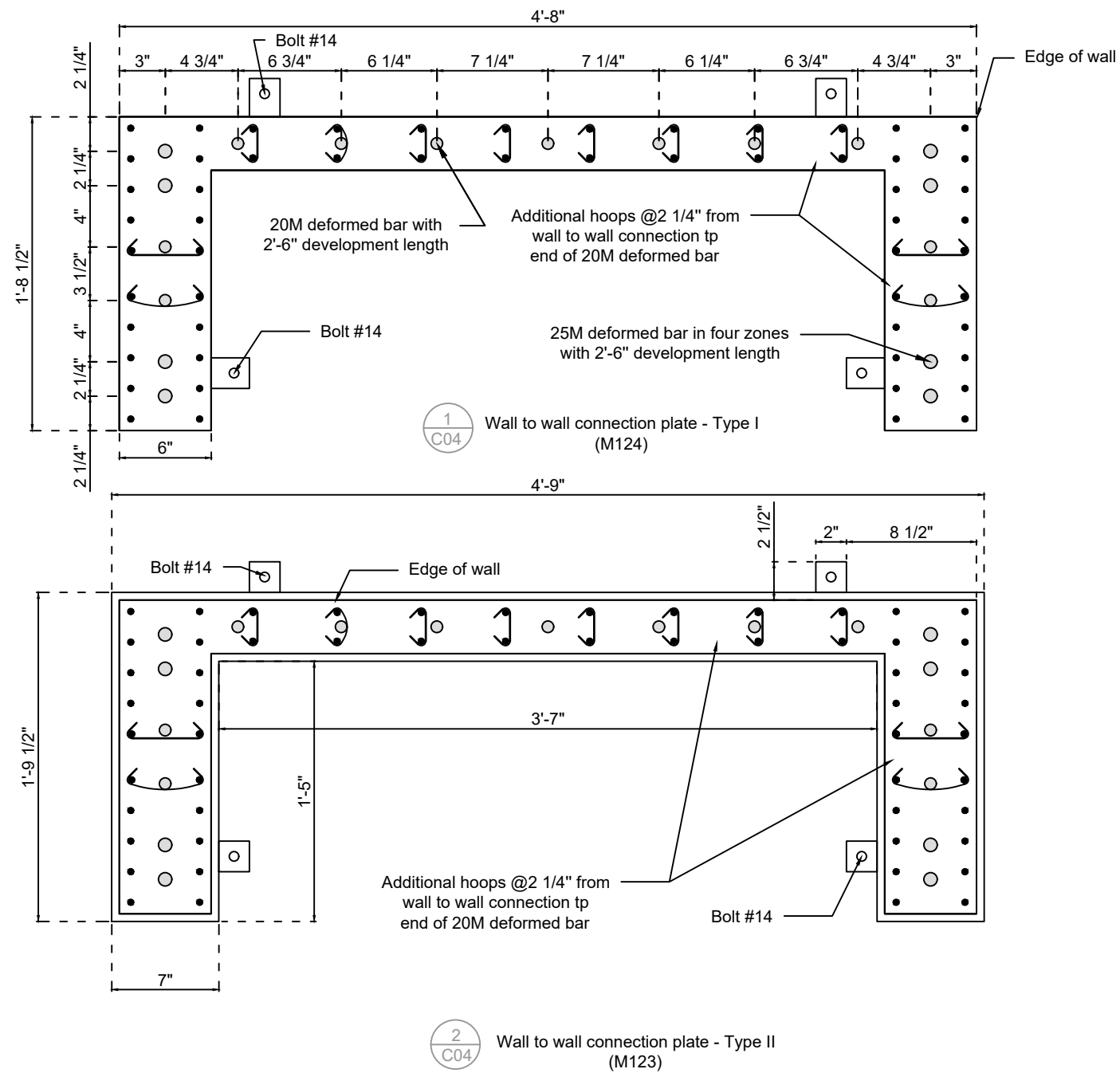
Special case with embed beam
(B111)

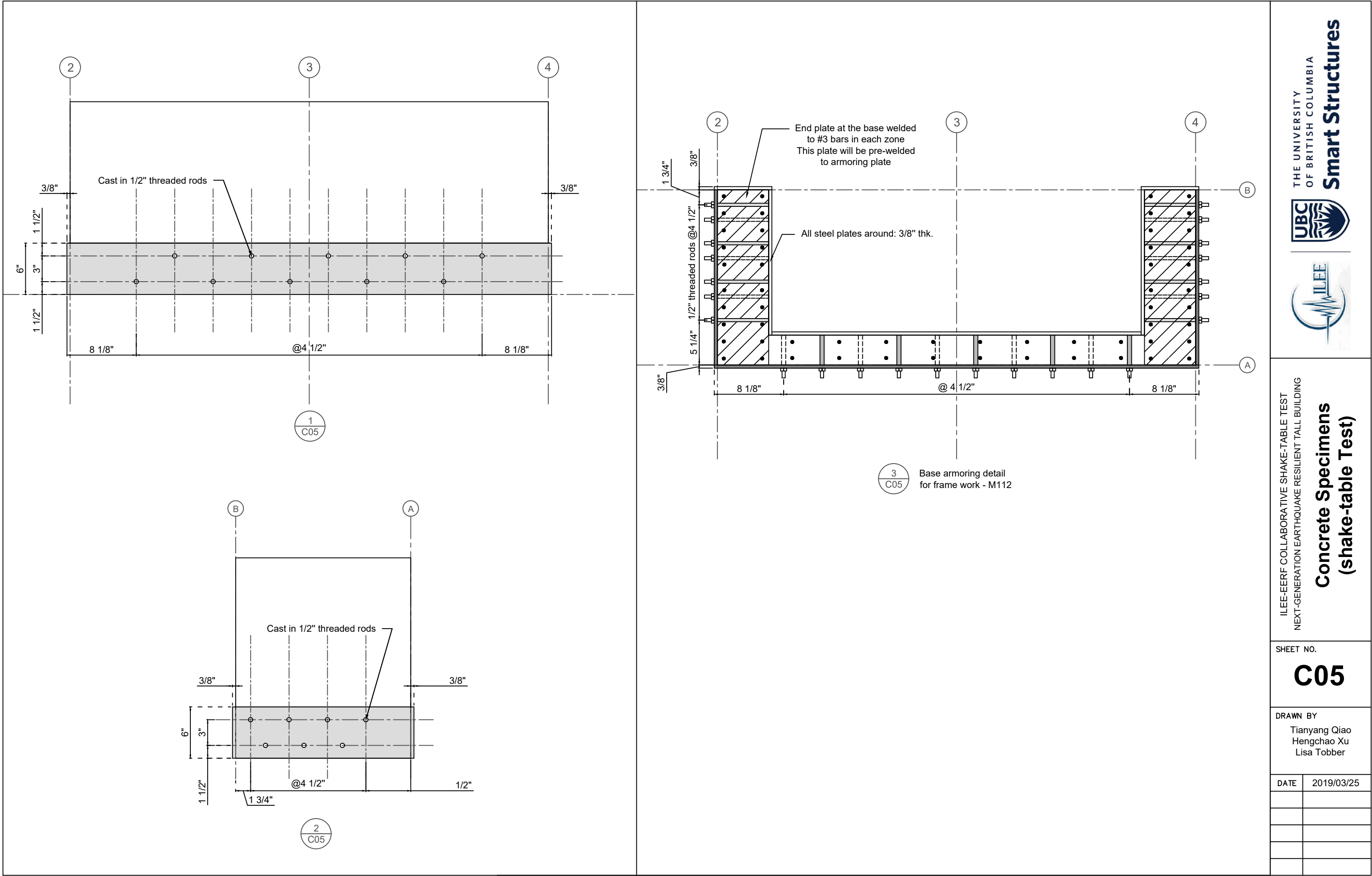


2
C03 Location of four zones, two flanges
and one web

LVL	Zone reinforcing		Distribution steel		f' _c
	Z1	Z2	SW1 (flange)	SW2 (web)	
Roof					
L24					45MPa
L23					
L22					
L21		8-#3 vert 11ga ties@2 1/4"			
L20					
L19					
L18					
L17					
L16					
L15					
L14					
L13					
L12					
L11					
L10					
L9					
L8					
L7					
L6		8-#3 vert 11ga ties@6"			
L5					
L4					
L3					
L2					
L1	8-#3 vert 11ga ties@2 1/4"	8-#3 vert 11ga ties@2 1/4"	6in wall #3 @ 3" VEF #3 @ 6" HEF	3.5in wall #3 @ 5 1/2" VEF #3 @ 6" HEF	

5
C03 Detailed information of reinforcement in elevation





ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Concrete Specimens
(shake-table Test)

SHEET NO.

C05

DRAWN BY
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Hengchao Xu
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DATE 2019/03/25

